MTRX3700 2026

Major Project

Group Assignment (groups of four)

Pitch video: due **Tuesday 13 October**, 11:59 pm; pitch in the lecture on **Wednesday 14 October**

Demonstration and individual Q&A: Week 13, **Thursday 5 November**; demonstration video **Wednesday 4 November**, 11:59 pm

Report and code: Sunday 8 November, 11:59 pm

Assessment: Pitch 3%, Presentation & Demonstration 8% and Report 8% (group). Individual Q&A 5%. Total 24%.

1. Introduction

After Piano Tiles and PNR-Sing, PNR-Play Incorporated wants a robot controlled by an FPGA. Build, test and explain a robot whose movement responds to its live camera. Your marks reward engineering achievement and evidence, not simply the number of features or sensors.

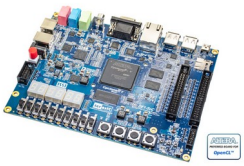
Pitch: specify the design and measurable goals. Presentation & demonstration: explain the result and show it working. Individual Q&A: explain your own modules/code that you contributed. Report: document the design, tests and measured performance.

Use Lessons 3 to 5, Assignment 2 and the Major Project Resources (to be released on Ed).

2. Task Specification

2.1 Robot kit and the provided hardware test

Use your DE1-SoC and the provided UGV02 robot kit (Figures 1 to 4). It includes the chassis and charger, Robot Adaptor Board, OV7670 camera, RCWL-1601 ultrasonic sensor/s (3.3 V only), IR remote, Arduino Nano 33 IoT, microphone, VGA cable and VGA–HDMI adaptor. Mount the DE1-SoC using the supplied rails. Pin assignments and hardware test code will be released in the Major Project Resources on Ed.



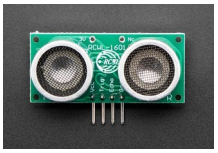
DE1-SoC



UGV02 chassis



OV7670 camera



Ultrasonic sensor (RCWL-1601, 3.3 V)



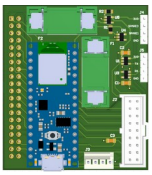
IR remote



Arduino Nano 33 IoT



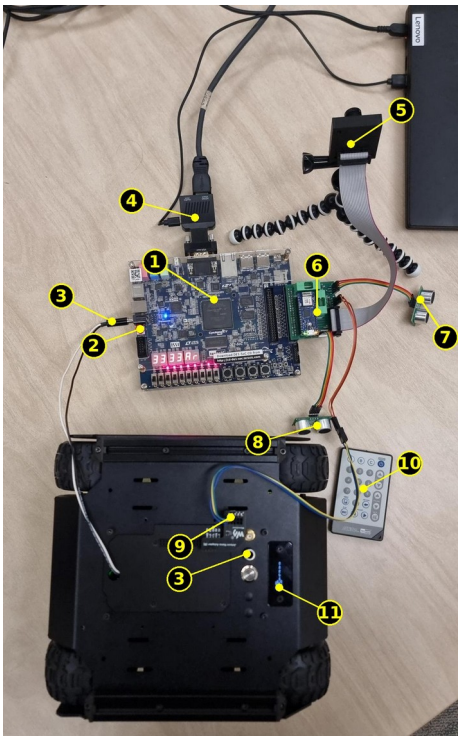
Microphone



Robot Adaptor Board (Arduino Nano 33 IoT fitted)

Figure 1: Your DE1-SoC and the parts of the robot kit.

Connect the complete kit as shown below. Check the ultrasonic polarity before power-up: a reversed sensor can easily be destroyed.



1 DE1-SoC. The Robot Adaptor Board (6) is on its GPIO_1 header, the one nearest the board's corner.

2 DE1-SoC power button (red). Press it after the chassis is on.

3 12 V lead from the chassis's DC socket to the DE1-SoC's power jack.

4 VGA cable to the monitor, here through the VGA–HDMI adaptor, which needs USB power from another device (a laptop in this example).

5 OV7670 camera on its ribbon, with the keyed plug in J2 on the Robot Adaptor Board.

6 Robot Adaptor Board, with the Arduino Nano in its socket, USB connector towards the DE1-SoC's buttons. Both fuses are pressed into their clips: F1 (5 V, for the Nano) and F2 (3.3 V, for everything else).

7 Ultrasonic sensor 2 on J4 (SPARE): VCC on the pin marked 3V3, GND on GND.

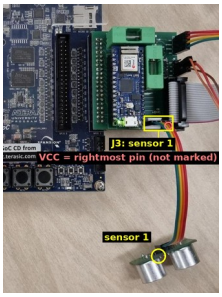
8 Ultrasonic sensor 1 on J3. J3's pins are not marked: with the board placed as in Figure 3(a), VCC is the rightmost pin, then TRIG, ECHO and GND.

9 Three wires from J5 to the chassis header: GND to GND, TX to RX, RX to TX. Never connect J5's 3V3 pin.

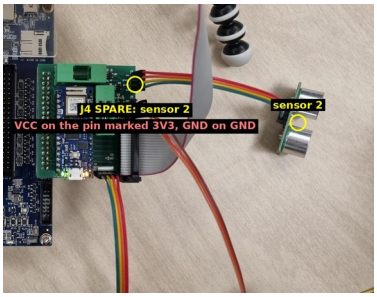
10 IR remote. Point this towards the IR receiver on the DE1-SoC's bottom-right corner (below the GPIO headers).

11 Chassis power switch and screen. The screen shows the battery voltage.

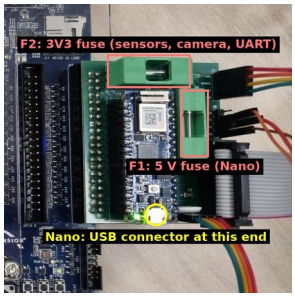
Figure 2: The kit wired for the hardware test.



(a) Sensor 1 on J3



(b) Sensor 2 on J4 (SPARE)



(c) Robot Adaptor Board: fuses and the Nano

Figure 3: Connections to check before power-up.

Run the supplied hardware test when you first receive the kit; LEDR0–LEDR7 lit confirms success (Figure 4). Report faults immediately before leaving the lab.

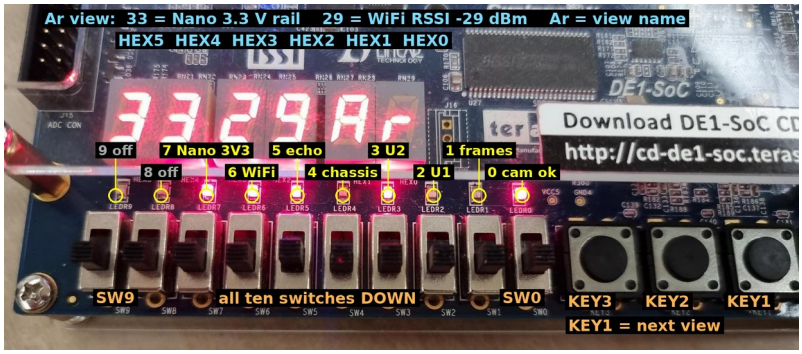


Figure 4: The hardware test passing. Each label names what its LED tests.

Practicalities:

- Secure cameras and sensors using removable or 3D-printed mounts.
- NEVER remove or modify the protective Plexiglas cover on the FPGA board.
- Keep the hardware portable so you can take them home (please clear them from the lab).
- Discuss extra components, connectivity and anything that could be hazardous with tutors before use.

2.2 Requirements (R1–R9)

ID	Requirement	How it is demonstrated
R1	Camera in the loop (**essential**)	Motion responds to a quantity computed on the DE1-SoC from live camera pixels. Displaying video alone is insufficient.
R2	FPGA-driven chassis (**essential**)	DE1-SoC commands the UGV02 by UART, including direction, turning and variable speed.
R3	Safety stop	IR remote stops the robot immediately from any state; it stays stopped until another IR remote key releases it.
R4	On-board processing	Sensor processing and motion decisions run on the DE1-SoC. Per-pixel/audio-sample work uses RTL; slower decisions may use a processor on the DE1-SoC. Arduino Nano is only a communications relay. Off-board devices may display/log but not replace processing, e.g. a laptop running ML on images sent from the DE1-SoC is allowed but does not satisfy R4.
R5	Measurable targets	Define 3–5 targets with measurable quantities, units, conditions, tests, threshold (least acceptable) and goal values; include at least one timing and one accuracy/repeatability target. See 2.3.
R6	Automated tests	run_all_tests.sh runs self-checking module tests and a system-level simulation in Verilator 5.050 (on your laptop or in an Ed workspace), prints PASS/FAIL, detects one deliberately broken module (prints CAUGHT for it) and finishes within 5 minutes from a clean copy.
R7	Golden model and replay	Model at least one self-designed processing module and compare its RTL with the model on recorded real-hardware inputs, e.g. in a Python notebook. Marked in demo Part B and report Q4.
R8	Ownership	OWNERS.md lists every source file, its SID owner and origin. Each member owns a new or adapted (more than changed parameters) synthesisable module and a testbench.
R9	Bypass test	Your main self-designed module/s must make a measurable difference. Ensure that a switch on the DE1-SoC disables them while the rest of the robot keeps running; the tutor repeats their target/s with them disabled. If the robot still meets the threshold without them, your level is set as if they were absent.

R1 and R2 must both work on the robot to reach Pass. R3 must work before the robot can be tested. Violations of R4 must be disclosed, and missing evidence of on-board processing loses the associated marks.

2.3 Measurable targets: examples, not prescribed performance

A target turns “it works” into a number the tutor can experimentally check in the demo. Each group must propose its own measurable performance targets and clearly communicate them during the demonstration. You need three to five targets, with at least one timing figure (a latency or a rate) and at least one accuracy or repeatability figure. Targets must reflect your own contribution, rather than functionality already provided.

A target has five parts:

1. **Quantity**, with its unit: what is measured.
2. **Conditions**, stated as ranges. On the day, the tutor chooses within them.
3. **Test**: how the tutor should measure the quantity. Use three trial iterations and specify which result is used out of the three (do not simply choose the best result out of the three).
4. **Threshold**: the minimum (or maximum) quantity for the target to be considered achieved.
5. **Goal**: the range of quantity values you are aiming for.

Choose targets appropriate to your robot and your own design. The examples below illustrate how to define acceptance tests; they are NOT required numerical specifications.

	Quantity (unit)	Conditions (the tutor chooses within)	Test	Threshold	Goal
T1	Wall stopping error (mm): the difference between the gap the robot leaves at a wall and its stated 300 mm	Approach speed 0.2–0.3 m/s; start 1–2 m from the wall	Three approaches; the tutor measures each gap with a tape measure; the largest error of the three iterations counts	≤ 50 mm	≤ 20 mm
T2	Camera-to-wheel command latency (ms): from the first frame that shows a change to the first UART bit of the command that responds	A change the tutor makes in the camera's view (e.g. moving the object the robot follows), where and when they choose	Three changes; the tutor sees each latency in a SignalTap capture or in UART timestamps; the longest of the three iterations counts	≤ 100 ms	≤ 50 ms
T3	End-position spread over repeated manoeuvres (mm): the largest distance between two end positions	The stated manoeuvre (e.g. 1 m forward, then a 90° turn), from a start mark and heading the tutor picks	Three runs; the tutor marks each end position on the floor; the largest distance between two marks counts	≤ 100 mm	≤ 40 mm

Design the test for the room. The tutor must be able to witness its three trials in about a minute and a half, with what **you bring** (a tape measure, floor tape, a laptop with SignalTap). It has three trials, run under conditions the tutor chooses within your ranges. A target is badly designed if:

- the provided code or the chassis already meets it without your work (such a target scores 0);
- it is scored on the best of several trials;
- it states no conditions;
- nobody can measure its quantity in the room.

Where targets go. For the demonstration, list them on the **one-page datasheet,** each with its predicted value and its measured value. Bring the datasheet to the demonstration, include it in your code upload (section 7), and reproduce it as Q1 of the report. The tutor tests your targets as they stand on that datasheet, which also names the level you claim (section 4). During the demonstration, present your predicted performance (from simulation or calculation), demonstrate the actual result, and explain any differences.

2.4 Achievement progression

Level	What the robot achieves	Evidence expected
Below Pass	Camera does not affect movement or the FPGA does not drive the chassis.	R1 or R2 not demonstrated.
Pass	The provided code plus the minimum glue that closes the loop (e.g. threshold → centroid → steering): reliable basic camera-guided movement, e.g. follow a visually detected region and stop safely.	R1–R8; the robot works from camera to wheels on repeated runs, and its targets are measured.
Credit	One processing stage beyond that minimum, designed by the group, that measurably improves a target (e.g. connected-component labelling instead of the bare centroid); the bypass switch shows the difference.	Self-checking testbench, golden model, hardware replay, and a measurable improvement with the bypass switch toggled (R9).
Distinction	A designed pipeline of several stages working in the closed loop, with a target that measures it (e.g. marker tracking with a predictor through occlusion).	Closed-loop demonstration, integration testing, target for the new feature, predicted timing/resources.

Level	What the robot achieves	Evidence expected
High Distinction	Something no provided material, or routine combination of it, does, with the simulation-to-hardware gap explained (e.g. visual odometry).	Quantitative comparison, repeatability, simulation-to-hardware analysis and justified design trade-offs.

Pass: the provided code plus the minimum glue that closes the loop. **Credit**: one stage beyond that, designed by you, that measurably improves a target. **Distinction**: a pipeline of several stages you designed. **High Distinction**: a capability that nothing provided has.

The Lookbook (to be released) gives a worked example for each level and places other ideas against them, including sensor fusion and the optional CNN lesson's network (the CNN lesson teaches what you need; no level requires one). Using more of the robot hardware kit's sensors is not by itself an achievement.

The tutor confirms the highest level actually demonstrated. A proposed higher level must be supported by a new/adapted module and measurable evidence. Reused staff-provided code is permitted but does not itself constitute an original contribution.

Level	Below Pass	Pass	Credit	Distinction	HD
Factor	25%	70%	80%	90%	100%

The factor multiplies the group Demo (8%) and Report (8%) marks only; Pitch (3%) and individual Q&A (5%) are unaffected. Targets are judged on their measured performance, not just ambition.

2.5 Implementation and verification

Decide what the robot does, which sensors complement the camera, image/frame formats, controller, timing, failure behaviour and hardware/software split. Simulate first with the device models that will be supplied with the Major Project Resources. Parameterise frame sizes/timing for fast tests. Reset before tests (ModelSim may start at X while Verilator starts at zero). Verify FPGA ALM, M10K, DSP, Fmax, slack and latency in Quartus.

Staff-provided code and your own earlier assignment code may be reused with attribution in OWNERS.md. External open-source code requires attribution; another group's code is not permitted. The owner must understand every submitted line, including AI-generated code.

3. Pitch [3 marks]

Submit GroupNumber_PITCH.mp4 (e.g. Group07_PITCH.mp4; up to **2 min 30 s**; H.264 where possible, 720p/1080p, up to 150 MB) to the designated Google Drive folder by **Tuesday 13 October, 11:59 pm**. Test the final file before uploading it: make sure both the video and audio play correctly. Your video plays at the **14 October lecture**, followed by about **1 min 30 s** of live questions and feedback. The video is the only pitch deliverable.

All group members must speak in the recorded video and be present during the pitch session. When it is your group's turn, the whole group comes to the front of the lecture theatre before the video starts and stays there for the questions and feedback. A first-come, first-served booking sheet, similar to A1's, will be released for you to select your presentation slot. With 25 groups presenting within the two-hour lecture, keep strictly to the allocated timing: 2 min 30 s video + 1 min 30 s questions and feedback per group.

The pitch should contain these four items, in about the suggested times. The right-hand column gives examples, not extra requirements.

Item (suggested time)	What a strong pitch shows
Goals of the project, emphasising achievable technology (20 s)	For example: robot purpose, the main new capability and feasibility.
Preliminary system design and expected technical requirements (60 s)	For example: a block diagram (camera → computation → wheels); the measurable targets you expect to meet, with threshold, goal and test (section 2.3).
Project roadmap and implementation feasibility (60 s)	For example: simulation plan, date of first camera-to-wheel test, main risks and fallbacks.
Distribution of tasks across your team (10 s)	For example: each member's module responsibility.

4. Recorded Presentation and Live Demonstration [8 marks]

Each group has 15 minutes: a recorded **5-minute** presentation with slides followed by a 10-minute live robot demonstration. Use the same video style as the pitch (up to 300 MB for the 5-minute video). Submit GroupNumber_DEMO.mp4 to the designated Google Drive folder by **Wednesday 4 November, 11:59 pm**. All group members contribute to the presentation and attend the demonstration. Individual Q&A is assessed separately.

As the slot begins, with the tutor watching, start run_all_tests.sh from a clean copy of the code you are demonstrating. It runs while the recorded presentation plays, and you read its result in Part B.

Part	What must be shown	Marks
A. Recorded presentation (5 min)	Robot purpose and system diagram; design decisions and ownership; datasheet and target results; test coverage, limitations and improvements.	2.0
B. Live verification (about 2 min)	Read the result of run_all_tests.sh (started as the slot began) and explain selected tests; show the golden model and replay against recorded hardware inputs, beside the board's result.	2.0
C. Live acceptance (about 7 min)	Safety stop (R3) first, then a complete run: the robot doing everything the group designed it to do, start to finish, as the datasheet describes. Then the tutor checks R1, R2 and R4 and selects three measurable targets to test; bypass run where applicable (R9).	3.5
D. Limits (about 1 min)	Explain an observed shortfall and its likely cause.	0.5

Groups must demonstrate the robot's full functionality. Bring a charged & assembled robot, the code and your one-page datasheet that names the grade level you claim and the module/s it rests on (section 2.4); the three targets tested should include the one/s that measure those module/s. The tutor chooses the test conditions within the datasheet's ranges: three witnessed trials per target, scored by the result its test specifies (never the best of the three). Trials may come from the complete run where the target's test allows it. Part C gives 0.5 for the complete run and 1 mark per tested target: in full at its goal, three-quarters at its threshold, half if it misses the threshold by less than the goal-to-threshold gap and you explain why, and nothing otherwise (including if it is not run). If the safety stop fails, no driving test is run. Marks reflect actual evidence, not an untested claim.

5. Individual Q&A [5 marks]

Each student answers questions about the files they own in OWNERS.md and their interfaces (approximately 5 minutes). AI-generated code must be understood by its owner.

Question	Expected demonstration
Explain	Find relevant code and explain its hardware behaviour.
Why	Justify a design decision and alternative.
What if	Predict the effect of changing a parameter or input.

6. Report (maximum 6 pages) [8 marks]

One group PDF on Canvas, due **Sunday 8 November, 11:59 pm.** Use SID only (no names); the contributions table and the one-page AI/decision Statement are outside the page limit. Figures must be labelled and legible.

Section	Required content	Marks
Q1 Datasheet	Robot summary, diagram, goal/threshold/predicted/measured values, conditions and limitations.	1.5
Q2 System design	Modules and interfaces, widths, clock domains, FSM, hardware/software split.	1.5
Q3 Design decisions	Two choices, rejected alternatives and quantitative reasons.	1.5
Q4 Verification	Regression and coverage, golden model/replay, deliberately broken module detected by test.	2.0
Q5 Hardware budget	Predicted/actual ALM, M10K, DSP; Fmax/slack and camera-to-wheel latency.	1.0
Q6 Debugging	Two genuine bugs: symptom, cause, fix and added regression test.	0.5
Q7 Contributions	SID ownership table and code uploader.	Required

7. Code submission

One member uploads code to the Major Project submission, which will open on Ed, by **Sunday 8 November, 11:59 pm**. Include Quartus project files (.qpf, .qsf, .sdc), RTL (.v, .sv), testbenches, software sources, the final .sof bitstream, run_all_tests.sh, OWNERS.md, golden models and replay data, and your one-page datasheet (PDF). Use relative file paths; omit regenerated db/, incremental_db/ and output_files/ folders, except the final .sof.

Please also upload a draft version of the code to Ed before the demonstration and individual Q&A. The draft version is used to prepare the tutors' questions for the Q&A (your OWNERS.md should be accurate).

8. Submitted Work AI Policy

AI is allowed for the preparation of the pitch, the report and any other unsupervised development, but not as assistance during the supervised demonstration or individual Q&A. You are responsible for understanding, testing and acknowledging AI-generated work. If you used AI, submit exported relevant AI sessions as separate files in the same Canvas submission as the report, and include a one-page Statement with tools, representative prompts, an error or limitation found and how outputs were verified. If you did not, include a one-page design decision and validation statement. The Statement is required and not marked.

9. Group work, safety and equipment

All members must contribute and complete SparkPlus. Group marks may be moderated using SparkPlus, OWNERS.md and individual understanding. Follow laboratory rules, keep the robot safe and portable, and return the complete kit in good condition in the week after the demonstration (9–13 November). University late-submission, special-consideration and academic-integrity policies apply. It is recommended that you keep an engineering lab book of calculations, issues and test results.